

4. Given an array which consist of only 0,1,2 sorting an array using sort algo

```
class Solution {
public:
    void sort012(vector<int> &arr) {
        int low = 0;           // position for next 0
        int mid = 0;           // current index
        int high = arr.size() - 1; // position for next 2

        while (mid <= high) {
            if (arr[mid] == 0) {
                swap(arr[low], arr[mid]);
                low++;
                mid++;
            }
            else if (arr[mid] == 1) {
                mid++;
            }
            else { // arr[mid] == 2
                swap(arr[mid], arr[high]);
                high--;
            }
        }
    }
};
```

Output:-

The screenshot shows a coding platform interface with a dark theme. On the left, the 'Output Window' is open, displaying 'Problem Solved Successfully' with a green checkmark. Below this, it shows 'Test Cases Passed: 1111 / 1111', 'Attempts: Correct / Total: 1 / 1', 'Accuracy: 100%', 'Points Scored: 4 / 4', and 'Your Total Score: 17'. On the right, the C++ code for the 'sort012' function is displayed, which implements the Dutch National Flag algorithm. The code is as follows:

```
1 class Solution {
2 public:
3     void sort012(vector<int> &arr) {
4         int low = 0;           // position for next 0
5         int mid = 0;           // current index
6         int high = arr.size() - 1; // position for next 2
7
8         while (mid <= high) {
9             if (arr[mid] == 0) {
10                 swap(arr[low], arr[mid]);
11                 low++;
12                 mid++;
13             }
14             else if (arr[mid] == 1) {
15                 mid++;
16             }
17             else { // arr[mid] == 2
18                 swap(arr[mid], arr[high]);
19                 high--;
20             }
21         }
22     }
23 };
24
```